

SCIENTIFIC SUMMARY · SCIENTIFIC WRITING & EVIDENCE SYNTHESIS

Identification of Host Endotypes Using Peripheral Blood Transcriptomics in a Prospective Cohort of Patients With Endocarditis

IN PLAIN TERMS — *two distinct immune signatures found in blood samples predict which endocarditis patients are most likely to survive.*

Summary prepared by Israel David Duarte-Herrera, first author of the original study

Published in International Journal of Infectious Diseases, 2024

Duarte-Herrera ID, López-Martínez C, Rodríguez-García R, et al. Identification of host endotypes using peripheral blood transcriptomics in a prospective cohort of patients with endocarditis. Int J Infect Dis. 2024;148:107235. DOI: [10.1016/j.ijid.2024.107235](https://doi.org/10.1016/j.ijid.2024.107235) · Open access, CC BY-NC-ND 4.0

§1 Executive Summary

AT A GLANCE

Endocarditis kills more than a third of the patients who develop it, and for years the only tools doctors have had to judge who's most at risk are the usual suspects — age, comorbidities, which valve is involved. We wanted to know if the immune system itself was already giving us a better answer, if we just knew how to read it.

We sequenced RNA from blood drawn at diagnosis in 32 patients and the transcriptomes split cleanly into two groups. One leaned on neutrophils and interferon signaling, the kind of fast, blunt response you'd expect in a systemic inflammatory reaction. The other leaned on T cells, a slower and more targeted response. Nothing on the chart gave this away — age, comorbidities, even the bacteria involved looked the same in both groups. Survival didn't: 56% of patients in the inflammatory group died in hospital, against 14% in the other. We saw the same split, and roughly the same survival gap, when we ran the same signature on blood from COVID-19 and sepsis patients, which is what convinced us this isn't an endocarditis-specific quirk but a broader pattern in how the body handles a severe infection.

None of this changes bedside practice tomorrow. But it suggests a blood test, rather than a symptom checklist, could eventually help flag which endocarditis patients need closer watching — and maybe which ones shouldn't be rushed into surgery.

32

endocarditis
patients profiled

2

host endotypes
identified

**56% vs
14%**

in-hospital
mortality, EE1 vs
EE2

2

external cohorts
confirming the
pattern

§2 Technical Summary

Background & Rationale

Two patients can show up with the same infected valve and the same organism, and one sails through treatment while the other ends up in the ICU. That gap usually gets chalked up to how sick someone "looks" on admission, but work in sepsis, ARDS, and severe COVID-19 has made a good case that host biology, not just the pathogen, drives a lot of that variability. These subgroups — defined by shared molecular mechanisms rather than shared symptoms — are what the field calls endotypes. Nobody had gone looking for them in endocarditis before this study.

Design & Methods

We ran a single-center, prospective cohort at Hospital Universitario Central de Asturias, enrolling 32 consecutive adults with definite endocarditis (Duke criteria) over a one-year window. Blood went for RNA-seq at diagnosis, and again the day after surgery for patients who had one. We clustered patients with Ward's method on the most variable genes and visualized the split with UMAP, estimated immune cell composition by deconvolution, and used GSEA to pull out which pathways were actually driving the separation. To see whether any of this held up outside endocarditis, we built a 20-gene score from the genes that differed most between the two groups and tested it — blind to the original clustering — on two cohorts we already had sitting in the freezer: 56 patients with severe COVID-19 and 19 with sepsis.

Key Results

FEATURE	ENDOTYPE 1 (EE1, N=18)	ENDOTYPE 2 (EE2, N=14)
Dominant biology	Interferon signaling, activated neutrophils	T-cell maturation, adaptive/lymphocyte response
Baseline clinical profile	No significant differences between groups at diagnosis	

In-hospital mortality	56% (10/18)	14% (2/14)
Adjusted hazard ratio for death	13.8 (95% CI 2.7–71.4) for EE1 vs EE2	

Note: the source article's abstract reports a closely related but not identical adjusted hazard ratio (12.987, 95% CI 3.356–50); the figure above reflects the value detailed in the Results, Discussion, and Figure 4 of the full text.

The split held up even though nothing in the clinical picture — age, comorbidities, valve type, which bacteria were involved — predicted it. Surgery complicated the story: in EE2 patients who went under the knife, the post-op transcriptome drifted toward an EE1-like, more inflammatory state. And every death in the EE2 group happened after surgery. We can't prove cause and effect from five surgical patients, but it's hard not to wonder whether the operation itself works as a second hit in a group whose baseline immune response looked relatively benign.

The same signature separated high- and low-mortality patients in two completely unrelated conditions — COVID-19 and sepsis. That's the strongest evidence we have that this reflects something about how a given immune system responds to a severe infection, not something particular to endocarditis.

Clinical Implications

- Neither endotype showed up on standard risk scores or clinical variables at diagnosis — this is a biological signal you simply won't catch on a chart review.
- A neutrophil/interferon-heavy profile might be worth flagging for closer monitoring, or for the kind of immunomodulatory approaches already under study in sepsis.
- The post-surgical shift we saw in EE2 patients is hypothesis-generating rather than practice-changing right now, but it's a real question mark over surgical timing and perioperative immune support in this subgroup.

Limitations

- This is a small, single-center cohort (n=32). The two validation cohorts were collected before this study and weren't randomized against our signature, so treat the external validation as supportive, not confirmatory.
- We assigned endotypes from a single blood draw — a different clustering approach, or a second time point, might tell a somewhat different story. And since surgery wasn't randomized, we can't fully rule out indication bias behind the outcomes we attribute to endotype.
- We didn't include healthy controls, on purpose: the point was characterizing disease-related response, not comparing against baseline health.

Infective endocarditis

Transcriptomics

Host endotypes

Critical care

Precision medicine

This is an original summary written for portfolio purposes and does not reproduce the source article's text. Full study, figures, and statistical detail: doi.org/10.1016/j.ijid.2024.107235. Portfolio: israelddh.github.io